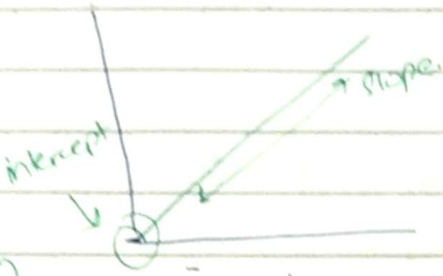
LAB 2

Linear Regression



① univariate linear regression
② Multivariate linear regression

Logistic Regression
→ falls in classification.
→ find output in class form (numerical : 0 to 1)

Polynomial Regression
→ Non-linear data

Univariate Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x$$

β_0 ← intercept value
β_1 ← slope

$$w_1 = \frac{N(\Sigma x_j y_j) - (\Sigma x_j)(\Sigma y_j)}{N(\Sigma x_j^2) - (\Sigma y_j)^2}$$

$$w_0 = (\Sigma y_j - w_1(\Sigma x_j))/N$$

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{1}{n}(\hat{y}_i - y_i)^2}$$

r^2_score = understanding of model of data
= ~~unders~~ accuracy of model pattern detection.

R^2 =

correlation is finded by the comparing the input of each feature by its output.

Multivariate Linear Regression:

$$h_w(x_j) = w_0 + w_1 x_{j,1} + \cdots + w_n x_{j,n} = w_0 + \sum_i w_i x_{j,i}.$$

Logistic Regression:

⇒ predicts value with categories.

⇒ sigmoid $1/1+e^{-z}$ e = euler numbers = 2.7

⇒ range 0 to 1

Polynomial Regression

fits the curved line by adding powers $x(x^2, x^3)$ etc